

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology & Engineering

Devang Patel Institute of Advance Technology and Research (DEPSTAR)

CSUE301 – : Big Data Analytics

Practical File – ODD Semester 2026-27

StudentID: 24DIT066

Student Name: Sil Shah

Practical 1

Aim: To learn basic Linux commands for system information, file and directory management, file searching, permissions, backups, and system monitoring in a Linux environment.

Problem definition:

You have joined **Data Analytics Inc.** as a Linux System Administrator. Your task is to set up the **Marketing Analytics** project environment by creating directories, managing files, monitoring system resources, searching data, and creating backups for the Marketing Analytics team.

Objectives:

- ❑ Identify the current user and working directory.
- ❑ Display Linux system information.
- ❑ Create and manage directories and files.
- ❑ Add and view file contents.
- ❑ Search files and text using Linux commands.
- ❑ Copy, rename, move, and delete files.
- ❑ Create backups and monitor system resources.
- ❑ Understand basic Linux file management operations.

Observation / Output

1. Identify the current user.

```
sil-shah@sil-shah-VMware-Virtual-Platform:~$ whoami  
sil-shah
```

2. Find the current working directory.

```
sil-shah@sil-shah-VMware-Virtual-Platform:~$ pwd  
/home/sil-shah
```

3. Display system information.

```
sil-shah@sil-shah-VMware-Virtual-Platform:~$ uname -a
Linux sil-shah-VMware-Virtual-Platform 7.0.0-28-generic #28-Ubuntu SMP PREEMPT_DYNAMIC Sun Jun 21 01:01:36 UTC 2026 x86_64 GNU/Linux
```

4. Check system uptime and date.

```
sil-shah@sil-shah-VMware-Virtual-Platform:~$ uptime
18:47:27 up 31 min, 1 user, load average: 0.46, 0.60, 0.52
sil-shah@sil-shah-VMware-Virtual-Platform:~$ date
Tue Aug 4 06:47:57 PM IST 2026
sil-shah@sil-shah-VMware-Virtual-Platform:~$
```

5. Create a directory named MarketingAnalytics

```
sil-shah@sil-shah-VMware-Virtual-Platform:~$ cd ~
sil-shah@sil-shah-VMware-Virtual-Platform:~$ mkdir MarketingAnalytics
sil-shah@sil-shah-VMware-Virtual-Platform:~$ ls
Desktop Documents Downloads MarketingAnalytics Music Pictures Public Templates Videos snap
sil-shah@sil-shah-VMware-Virtual-Platform:~$
```

6. Create subdirectories: RawData, ProcessedData, Reports.

```
sil-shah@sil-shah-VMware-Virtual-Platform:~$ cd MarketingAnalytics
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ cd MarketingAnalytics
bash: cd: MarketingAnalytics: No such file or directory
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ ls
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ mkdir RawData ProcessedData Reports
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ ls
ProcessedData RawData Reports
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$
```

7. Create data and report files.

```
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ cd RawData
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics/RawData$ touch customer_data.txt
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics/RawData$ cd ../Reports
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics/Reports$ touch report1.txt
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics/Reports$
```

8. Add sample content.

```

sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ cd RawData
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics/RawData$ touch customer_data.txt
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics/RawData$ cd ../Reports
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics/Reports$ touch report1.txt
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics/Reports$ cd ../RawData
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics/RawData$ nano customer_data.txt
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics/RawData$ cd ../Reports
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics/Reports$ nano report1.txt

```

9. Display file contents.

```

sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics/Reports$ cat report1.txt
Marketing Analytics Report

Sales increased by 18%

Prepared by Sil Shah
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics/Reports$ cat ../RawData/customer_data.txt
CustomerID,Name,City
101,Raj,Ahmedabad
102,Priya,Vadodara
103,Amit,Surat
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics/Reports$

```

10. List all files and directories.

```

sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics/Reports$ ls
report1.txt
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics/Reports$ ls -l
total 4
-rw-rw-r-- 1 sil-shah sil-shah 73 Aug  4 18:51 report1.txt
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics/Reports$ ls -R
.:
report1.txt

```

11. Search for report files.

```

sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics/Reports$ cd ..
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ find . -name "*.txt"
./Reports/report1.txt
./RawData/customer_data.txt
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ find . -name "report*"
./Reports/report1.txt

```

12. Search specific content inside files.

```

sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ grep "Sales" Reports/report1.txt
Sales increased by 18%
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ grep "Raj" RawData/customer_data.txt
101,Raj,Ahmedabad

```

13. Copy report files to Backup directory.

```

sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ mkdir Backup
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ cp Reports/report1.txt Backup/
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ ls Backup
report1.txt

```

14. Rename files.

```

sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ mv Reports/report1.txt Reports/Marketing_Report.txt
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ ls Reports
Marketing_Report.txt

```

15. Move files between directories.

```
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ mv RawData/customer_data.txt ProcessedData/  
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ ls ProcessedData  
customer_data.txt
```

16. Delete a file.

```
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ rm Backup/report1.txt  
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ ls Backup
```

17. Delete an empty directory.

```
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ rmdir Backup  
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ ls  
ProcessedData RawData Reports  
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$
```

Result :

The Linux project environment was successfully created. Various Linux commands were executed to manage files and directories, search data, perform backup operations, and monitor system resources successfully.

Key Questions /Analysis:

Q1. What is the difference between absolute and relative paths?

- Absolute Path: Starts from the root directory (/) and gives the complete location of a file or directory.
- Relative Path: Starts from the current working directory and depends on the current location.

Q2. Why are Linux permissions important for file security?

Linux permissions protect files from unauthorized access, control read, write, and execute permissions, and improve system security and data integrity.

Q3. What is the purpose of the grep command?

The grep command searches for specific words or patterns inside one or more files and filters command output.

Q4. How does Linux support efficient file management?

Linux provides a hierarchical directory structure, powerful file management commands (ls, cp, mv, rm, find, grep), file permissions, and shell scripting for automation.

Q5. Why are backups important in enterprise environments?

Backups prevent permanent data loss, help recover files after failures or accidental deletion, protect against malware, and support business continuity.

Supplementary Problems:

1. Create a shell script to automate project directory creation.

```
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ nano create_project.sh
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ chmod +x create_project.sh
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ ./create_project.sh
Project directories created successfully.
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$
```

2. Find the top 10 largest files in the system.

```
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ ./create_project.sh
Project directories created successfully.
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ find / -type f -exec du -h {} + 2>/dev/null | sort -rh | head -10
4.1G    /swap.img
607M    /var/lib/snapd/seed/snaps/gnome-46-2404_153.snap
395M    /var/lib/snapd/seed/snaps/mesa-2404_1165.snap
297M    /usr/local/hadoop/share/hadoop/tools/lib/aws-java-sdk-bundle-1.12.367.jar
274M    /var/lib/snapd/seed/snaps/firefox_8107.snap
169M    /snap/firefox/8107/usr/lib/firefox/libxul.so
162M    /var/cache/apt/archives/linux-modules-7.0.0-27-generic_7.0.0-27.27_and64.deb
155M    /snap/mesa-2404/1165/usr/lib/i386-linux-gnu/libLLVM.so.20.1
140M    /snap/mesa-2404/1165/usr/lib/i386-linux-gnu/libLLVM.so.19.1
137M    /usr/lib/jvm/java-11-openjdk-amd64/lib/modules
```

3. Schedule a backup task using cron.

```
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ crontab -e
no crontab for sil-shah - using an empty one
Select an editor. To change later, run select-editor again.
 1. /bin/nano        <---- easiest
 2. /usr/bin/vim.tiny
 3. /bin/ed

Choose 1-3 [1]: 1
crontab: installing new crontab
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ crontab -e
crontab: installing new crontab
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ crontab -l
0 18 * * * cp -r ~/MarketingAnalytics ~/MarketingAnalytics_Backup# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
```

4. Monitor CPU utilization continuously using top/htop.

```
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ top
```

top - 19:05:34 up 49 min, 1 user, load average: 0.58, 0.63, 0.52
Tasks: 357 total, 1 running, 355 sleeping, 0 stopped, 1 zombie
%Cpu(s): 6.4 us, 1.8 sy, 0.0 ni, 91.7 id, 0.0 wa, 0.0 hi, 0.2 si, 0.0 st
MiB Mem : 5360.6 total, 816.5 free, 1607.1 used, 3272.6 buff/cache
MiB Swap: 4096.0 total, 4095.4 free, 0.6 used, 3753.5 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
10524	sil-shah	20	0	4069856	364920	153088	S	28.2	6.6	1:23.09	ptxys
3869	sil-shah	20	0	6546652	385640	178388	S	27.9	7.0	5:57.97	gnome-shell
1277	root	20	0	264600	10936	9168	S	2.3	0.2	0:30.99	vmtoolsd
11542	sil-shah	20	0	22112	6360	4180	R	2.3	0.1	0:00.27	top
4106	sil-shah	20	0	980580	104660	89200	S	2.0	1.9	0:38.46	vmtoolsd
9229	root	20	0	0	0	0	I	1.6	0.0	0:05.13	kworker/7:0-events
10104	root	20	0	0	0	0	I	0.7	0.0	0:03.68	kworker/2:0-mm_percpu_wq
10954	root	20	0	0	0	0	I	0.7	0.0	0:02.35	kworker/4:2-events
11376	root	20	0	0	0	0	I	0.7	0.0	0:00.65	kworker/5:0-mm_percpu_wq
14	root	20	0	0	0	0	S	0.3	0.0	0:05.81	ksoftirqd/0
15	root	20	0	0	0	0	I	0.3	0.0	0:02.98	rcu_preempt
Trash	root	rt	0	0	0	0	S	0.3	0.0	0:00.15	migration/0
815	systemd+	20	0	16980	7668	6616	S	0.3	0.1	0:06.21	systemd-oomd
3539	sil-shah	20	0	10064	7008	4780	S	0.3	0.1	0:02.28	dbus-daemon
9858	root	20	0	0	0	0	I	0.3	0.0	0:04.13	kworker/1:1-events
9865	root	20	0	0	0	0	I	0.3	0.0	0:00.70	kworker/3:0-events
10988	root	20	0	0	0	0	I	0.3	0.0	0:00.65	kworker/u514:0-events_unbound
11170	root	20	0	0	0	0	I	0.3	0.0	0:00.49	kworker/u513:2-flush-8:0
1	root	20	0	26964	17684	11736	S	0.0	0.3	0:11.83	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.15	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
12	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker/u512:0-iov6_addrconf

5. Generate a disk usage report for all project directories.

```
sil-shah@sil-shah-VMware-Virtual-Platform:~/MarketingAnalytics$ du -sh ~/MarketingAnalytics/*
```

4.0K	/home/sil-shah/MarketingAnalytics/create_project.sh
20K	/home/sil-shah/MarketingAnalytics/MarketingAnalytics
8.0K	/home/sil-shah/MarketingAnalytics/ProcessedData
4.0K	/home/sil-shah/MarketingAnalytics/RawData
8.0K	/home/sil-shah/MarketingAnalytics/Reports

Tools /Technology used:

- ❑ Ubuntu Linux (WSL)
- ❑ Bash Shell
- ❑ Linux Terminal
- ❑ Nano Editor
- ❑ Cron
- ❑ Core Linux Commands (whoami, pwd, ls, mkdir, touch, cat, grep, find, cp, mv, rm, rmdir, du, top)

Conclusion:

This practical provided hands-on experience with essential Linux commands for system administration. It demonstrated how to create and manage files and directories, search data, perform backup operations, monitor system resources, and automate tasks using shell scripts and cron, building a strong foundation in Linux-based file management and administration.